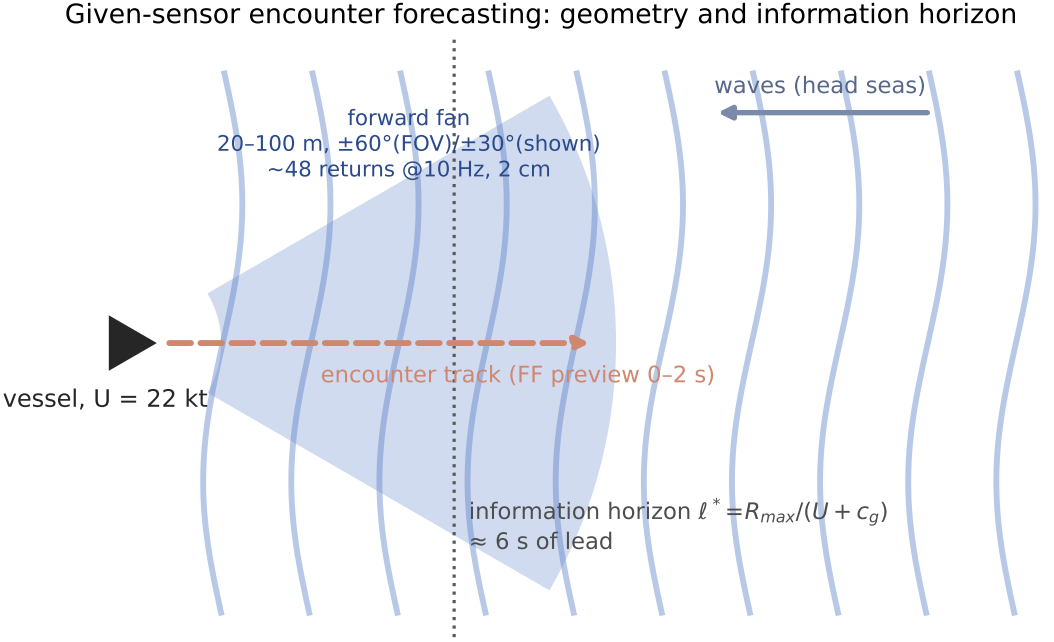
Given-sensor encounter forecasting: geometry and information horizon
waves (head seas)
forward fan
20–100 m, ±60°(FOV), ±30°(shown)
~48 returns @10 Hz, 2 cm
vessel, U = 22 kt
encounter track (FF preview 0–2 s)
information horizon $\ell^* = R_{max}/(U + c_g)$
≈ 6 s of lead